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Ballot Measure 110's effect on Crime: A Difference in Difference Approach

Introduction:

The issue of drug decriminalization is an often controversial one within the United States. Drug decriminalization challenges more traditional approaches to drug policy that prioritize punishment rather than rehabilitation. For the past half century, the United States has emphasized a heightened skepticism toward "weakness" in its discourse around crime and punishment. Being tough on crime was seen as a valuable characteristic in American politics. The "war-on-drugs" federal initiatives by the Nixon administration sought to bring a re-invigorated focus on the discouragement of the production, distribution, and consumption of a variety of drugs, which largely ushered in an era of prioritizing hard stances on drug offenders. This approach, however, has been widely criticized since its implementation for its failures in addressing drug addiction and its subsequent consequences.

Likewise, issues tangential to hardline drug policy such as mass incarceration and racial disparities have also gained increased attention and scrutiny in recent years, further highlighting the need for reevaluation and reform of existing drug policy approaches. Failures such as these have led policy makers to reconsider the stances of old as there has been a new paradigm shift away from traditional approaches to drug policy and toward decriminalization. Within the last

ten years, the United States has seen a wave of legalization for both the medicinal and recreational use of marijuana. Likewise, attitudes among people have changed around drug policy, with a greater emphasis on a public-health oriented approach to drug offenders. Rather than treating it as a criminal offence, drug addiction is treated as a problem that requires more compassion, support, and health intervention from its community members. Along with fixing the issues of hardline stances towards drugs, drug decriminalization is also often seen as a way to reduce additional harms associated with drug use, such as overdoses and infectious disease spread.

On the coattails of this newest paradigm shift in drug policy approaches comes Oregon's Ballot Measure 110, known as the Drug Addiction Treatment and Recovery Act. Passed in November of 2020, the new law aims to address many of the key issues with former drug policy stances. Instead of facing jailtime for possession of small amounts of drugs such as heroin, offenders can now either take a fine of 100 dollars or choose to receive a health assessment at a drug addiction treatment and recovery facility. The measure also establishes a mechanism for funding the program through the state's marijuana tax revenue and other savings from the reduction of arrests and incarcerations from drug related offenses. While the measure may inspire hope for future laws elsewhere, there are still plenty of critics of the new approach. For example, the state itself has released statements about the failures of its drug treatment facilities (Desaulnier, 2023). There is also the reoccurring concern about the measure's impact on public safety (Baertlein, 2022). This focus on public safety is the subject of my research. For decades, the "war-on-drugs" dominated our national discourse with fears over the impacts on public safety due to drug proliferation and decriminalization. The new law provides a great opportunity to test and understand that era-dominating assumption in a natural setting. Furthermore, it can help as a

critical tool in evaluating the effectiveness of a portion of the new law's impact on the state safety. This is important to emphasize, I will focus solely on the measure's impact on crime rate rather than other merits of the new law such as racial disparities and drug overdoses. To evaluate the full impact of the law on factors such as social justice, fiscal responsibility, and overall societal well-being, further research must be done.

Literature Review:

Drugs and crime are two interrelated issues that have been studied for decades, with research aiming to understand the linkages between them, as well as the potential impacts of drug policies on crime rates and criminal justice system. Drug use and addiction have brought justifiable concerns from stakeholders because of its connections to various forms of illegal activity.

These linkages were demonstrated through Goldstein's (1985) paper on the framework of drugs and crime. In his article, *The Drugs/Violence Nexus: A Tripartite Conceptual Framework*, Goldstein first addresses the psychopharmacological link of drugs in crime. This linkage occurs when people may commit crimes after the use of psychoactive drugs. The second link, economic compulsive, occurs when drug users commit crimes to acquire funds to spend on their drug use. And lastly, the third link is the systemic link, where motivations for profit within the market will produce violent outcomes due to competition and encourage overall involvement in additional crimes. These three links lay the groundwork for our theoretical discussion surrounding drug use and crime. They give the answer to broader questions as to why drugs and crime might be connected, however they leave out a lot of additional context.

The relationship between drugs and crime likely differs depending on the culture, geography, time factors, and the specific type of drug. This is something that Rolando et al (2020) demonstrate in their paper as a retort to Goldstein's framework. In their paper, they test the underlying linkages presented within Goldstein's paper and find that the relationships between drugs and crime are much more multi-faceted than presented in traditional research on the topic. Their research, based around interviews from young drug users that were involved with the criminal justice system in several different European countries, focused on the underlying mechanisms surrounding drugs and crimes. Throughout their interviews, the subjects demonstrated a multitude of reasons of why they offended. Firstly, there was some evidence which fell in line with Goldstein's economic compulsive linkage. Some respondents detailed how their need to purchase drugs led them to commit crimes for the money. Interestingly, this was the fact for both expensive and inexpensive drugs and also those more addictive such as heroin or cocaine and less addictive drugs such as marijuana. However, many noted their increase in aggressiveness when taking stimulants like cocaine, which led them to commit more crimes, an example of Goldstein's psychopharmacological link. Additionally, the researchers also found some nuance behind the lifestyle mechanisms of these offenders. There were many interviewees that demonstrated clear socioeconomic need which led them to their offenses rather than from the drugs themselves. Drugs could act to either "numb" some surrounding apprehension about committing crime or get the person in a more aggressive state for the same result but were not the main reason provided by those interviewed for their committing crimes. Researchers found this to be a common trend among those they interviewed, who said that they needed money for necessities or other items. Others pointed to the situation around them which brought them into the environment of crime. Social circle influences, as well as a culture which

values self-sufficiency was stated by several people as reasons for them to go expand into selling drugs. Crimes might have been drug-inspired at first, however the respect or money gained was said to have been the leading causes for any additional crimes. Research by Bennett and Holloway (2009) echo similar findings revolving around culture. In a similar study based around narratives of drug offenders, the researchers asked their respondents about the nature of their crimes and how drug use might have had connections to their criminal offenses. From their interviews, they conclude that Goldstein's taxonomy of crime and drugs leaves out the larger role culture plays in people's decision making surrounding the relationship between drugs and crime. Social group culture, they argue, influences individuals much more than the actual drugs themselves. Street culture, in particular, led individuals to want to involve themselves further in a desperate chase for money and satisfaction. Furthermore, cultural values surrounding money and spending also influenced the levels of crime, as people aspired to live an affluent lifestyle. Other reasons noted include a pursuit of hedonism and to "live life as party". These findings by Rolando et al and Bennet and Holloway, are important to keep in mind. Drug's relation to crime is affected by many different things, such as the discussed socioeconomic difficulties and cultural factors. When developing a model to understand drug decriminalization on crime, the inclusion of such factors is important for estimating drugs casual effects on crime as they control for additional crime relating to but not directly caused by drug use.

Moving from the theoretical linkages of drugs to crime, we see research empirically investigating how drug policy changes, in the form of decriminalization, have influenced crime in the real world. Research on this topic has been very promising for the side of decriminalization. Studies on marijuana legalization, for instance, have provided evidence towards a reduction or no significant difference in crime rates pre and post legalization. The

article, *The Implications of Marijuana Decriminalization and Legalization on Crime in the United States*, studies the impact of marijuana legalization across all 50 states between 2010 and 2014 (Maier et al, 2017). The authors find that, when controlling for factors leading to crime, there was no significant difference in violent or property crime depending on the status of marijuana legalization. Furthermore, research looking at marijuana decriminalization in Uruguay found similar evidence in the form of neighborhood safety (Soule, 2020). Both actual and perceived levels of safety within the neighborhoods studied remained similar to pre levels throughout the drug's legalization process and beyond. The optimism around drug decriminalization research doesn't stop at marijuana, similar research has been done on drug reforms passed in Tijuana, Mexico on drugs extending beyond marijuana. The reforms decriminalized small amounts of drugs and mandated diversion treatment for drug abusers. Researchers studying the effects of the change found no significant impact on violent crime after the city adopted the new law (Arredondo, 2018). These results give us some robustness for an attitude of optimism surrounding drug decriminalization. As the researchers are quick to point out, Tijuana is a major city for drug trafficking because of its proximity to the United States. Additionally, the city has a high concentration of drug users. Theory would suggest that the decrease in punishment and therefore greater incentive to consume drugs (especially for city known for its access to said drugs) would lead to greater number of offenses. Rather, this research once again shows a different result.

More broadly, research specifically looking at the effects of increases in drug incarceration on crime rates over time is more mixed. In their article, *The War on Drugs and Crime Rates*, Saadatmand et al find from 1970 to 2009, the war on drugs, measured using a proxy of percentage of the federal prison population, led to increases in total crime rate

(Saadatmand, 2012). However, when measuring the war on drugs as the spending on the Drug Enforcement Agency (DEA) by the federal government, the researchers do see significant negative effects on crime rate.

The relevant literature on the topic has a mixed tone surrounding the potential of Oregon's ballot measure on crime rate. On one hand, we have fundamental theory showing that drug decriminalization has a chance of leading to greater crime, as drug users and surrounding drug infrastructure contribute to crime, however muddy the mechanism behind it is (i.e., cultural factors, social circles, socio-economic status). On the other, we have a variety of real-world evidence pointing to the opposite conclusion. Considering the literature, this makes research into the ballot measure all the more helpful to our overall understanding of the topic of drug decriminalization.

Data:

The data for this paper was collected from the Oregon and Idaho State Police uniform crime reporting database for the date range of January 2020 to February 2023 with observations monthly. Data on socio-economic factors (unemployment rate and education level) comes from the U.S. Bureau of Labor Statistics and the U.S. Census and American Community Survey estimates. This data is also from the dates range of January 2020 to February 2023. The time interval chosen was due to Oregon's online reporting limitations, and the time relevance to the new law, as it was introduced in February of 2021. Our main variable of interest, crime rate, is composed of two main crime descriptions. One being crimes against persons, which include assaults, murders etc., which can be characterized by crimes whose victims are individuals. The other description is crime against property such as burglary and arson. These two types of crimes were combined to create my crime rate variable. Which in the model, is the main dependent

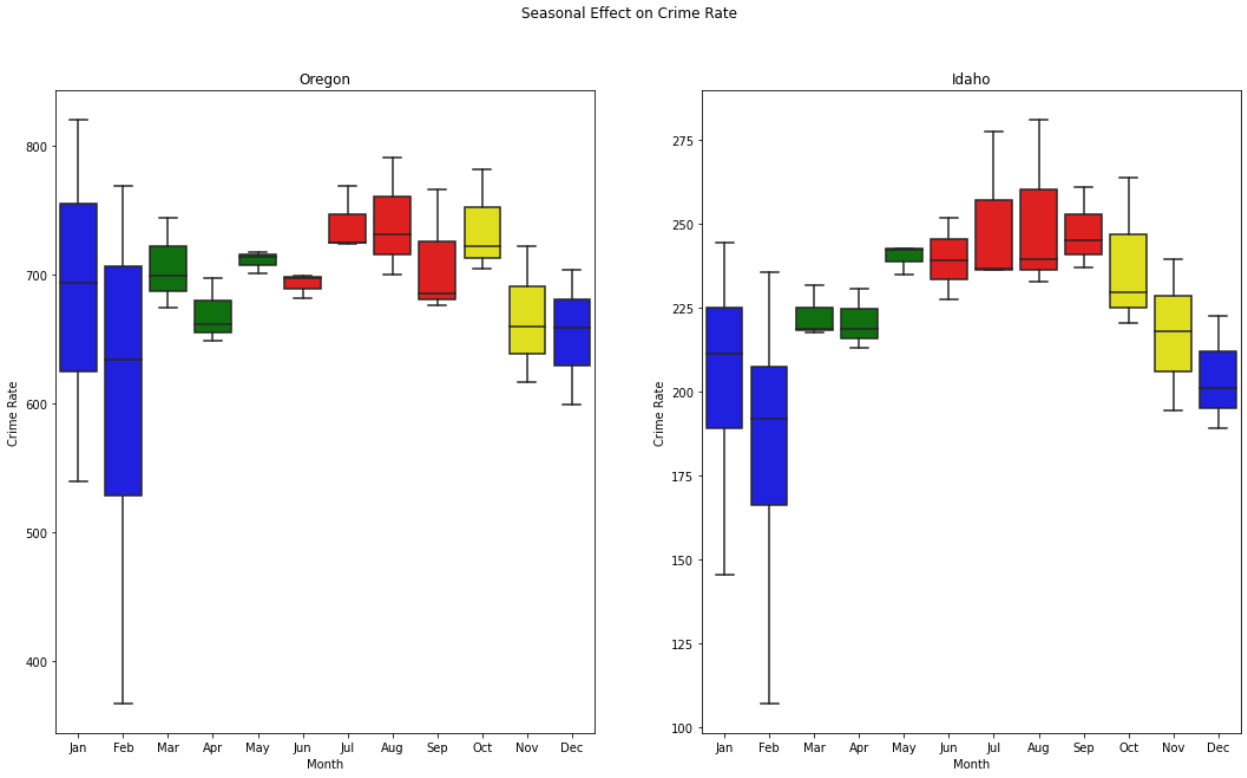
variable. I combined the two variables into one for simplicity in reporting and data analysis. Furthermore, a single variable relating to crime rate provides a more comprehensive measurement of the overall crime rate between the two states, providing a more holistic view of the crime environment in each given area. The addition of unemployment rate as a control variable is meant to capture some of the effect of socio-economic status on crime rate. Based on our literature review, we both theoretically and narratively know that socioeconomic factors may have an influence on a person's decision to commit a crime. Therefore, controlling the unemployment rate is a good way to capture the effects of variations in each state's socio-economic status. The table below describes the full list of variables used within our model.

Table 1: Description of variables

Name	Variable Label
Year	Year of observation
OR	=1 if Oregon
Crime Rate	Overall Crime Rate
Treatment	=1 if Treatment
Unemployment Rate	Unemployment Rate
Percent College	Percentage Some College or Above
February	=1 if February
March	=1 if March
April	=1 if April
May	=1 if May
June	=1 if June
July	=1 if July
August	=1 if August
September	=1 if September
October	=1 if October
November	=1 if November
December	=1 if December

The percentage of people with some college or above was also included as a control variable. This is because it captures both a part of socio-economic status and captures the differences in education levels between the two states. This is important because economic theory suggests that educational attainment has a negative effect on crime rate (Lockner, 2020). Differences, therefore, in Oregon and Idaho's levels of education would prevent us from

capturing the true effect of our dependent variable, Crime Rate, and introduce bias into our model. In addition to variables relating to socio-economic status, I included variables related to seasons. Based on preliminary research and exploratory analysis, weather and seasons seem to have effects on the level of crime. Table 2, for example, shows the distribution of crime rate depending on the month of the year between Oregon and Idaho in the years observed (2020-2023).



We can see that levels seem to spike in the warmer months and decline with colder ones. We also see small differences in seasonal crime rates between the two states. In Idaho crime rises higher in the summer compared to colder months relative to Oregon. Oregon, however, has colder months that maintain higher levels of crime relative to Idaho. To help control these seasonal differences in crime between the states, I included 11 different dummy variables corresponding to each month for which the crime data is connected to. The variable OR

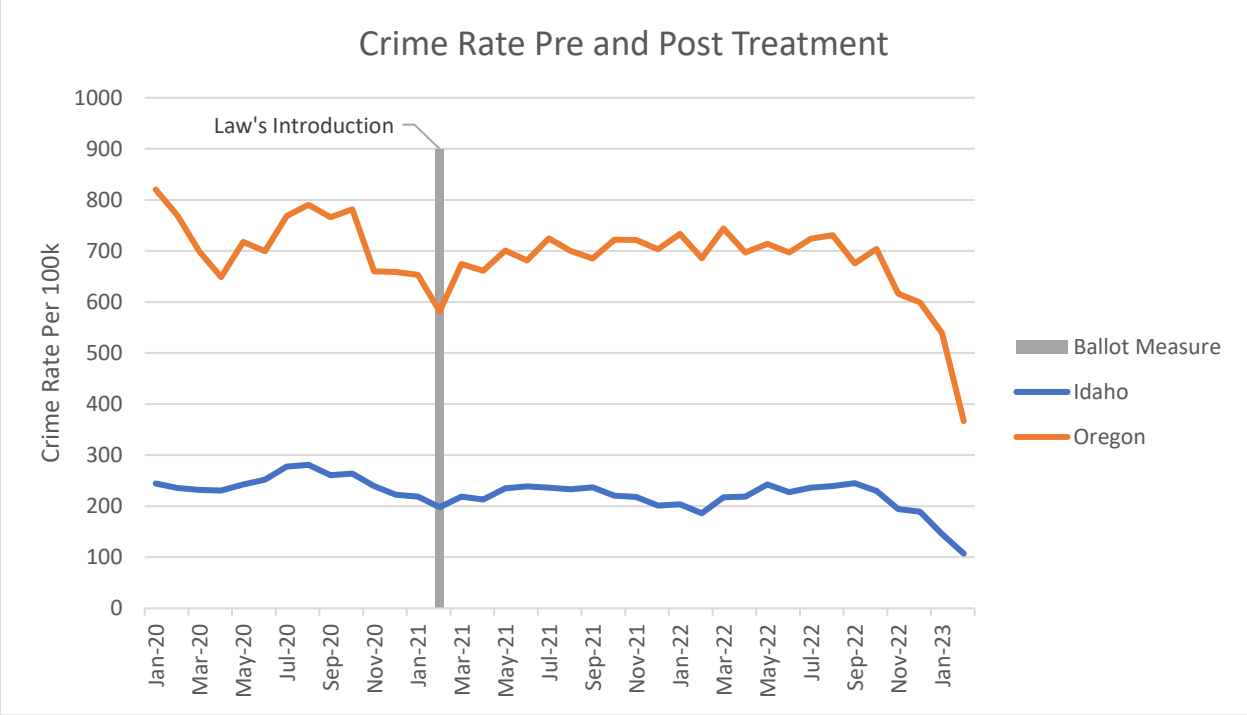
represents a dummy variable where 1 corresponds to an Oregon and 0 represents Idaho. Because we are working with a panel data set, these indicator variables let us differentiate observations between the states. The panel of data is also balanced between the two states, as demonstrated by mean average of the descriptive statistics table found below.

Table 2: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Year	76	2021.105	.918	2020	2023
OR	76	.5	.503	0	1
Crime Rate	76	457.238	241.597	107.115	820.271
Treatment	76	.658	.478	0	1
Unemployment Rate	76	4.718	2.319	2.5	13.7
Percent college	76	50.003	3.38	45.9	55.6
February	76	.105	.309	0	1
March	76	.079	.271	0	1
April	76	.079	.271	0	1
May	76	.079	.271	0	1
June	76	.079	.271	0	1
July	76	.079	.271	0	1
August	76	.079	.271	0	1
September	76	.079	.271	0	1
October	75	.08	.273	0	1
November	76	.079	.271	0	1
December	76	.079	.271	0	1

Likewise, the variable Treatment is an indicator variable denoting pre and post treatment time periods. Where the time periods (in months) are before (0) and after (1) the ballot measure was put into effect in February of 2021. Based on the mean average presented in Table 2, we see that we have an uneven number of treatment time periods to untreated time periods. In fact, around two-thirds of the data happen after the new law was introduced. While there is no requirement for them to be equal time periods before and after treatment for my model, the presence of this issue does make the difference-in-difference estimator noisier than it would be otherwise. Also, without more time periods it is difficult to tell if the parallel trends assumption

of the difference-in-difference model is violated or not via visual inspection. This may undercut any reliability in any casual effects I find via the results of the model. As shown the graph below, more data on the previous time periods would more clearly project the time trends of Oregon' crime rate, leading to a better assessment of the model's assumptions. However, this graphic does show that the two states did have similar trends in the albeit small amount of pre-treatment time periods. This is shown with movements up or down in similar months between the two states.



For the data, I also had to reckon with the possibility of measurement error from my data sources. It is clear from the graph above that there is a significant decrease in the crime rate as the time periods get closer to the present day. This could be from incomplete reporting by the state agencies from which I got the data. Throughout the writing process, I have checked repeatedly to see if any new data has become available for crime rate statistics and yet the same exists, with additional months added. While skeptical, the data is correct to the best of my knowledge and therefore will be used in the analysis.

Methods:

For this study, I used a difference-in-difference model with fixed effects. I felt the approach of using difference-in-difference made sense for an issue like this as it is able to differentiate the effects of the policy change from the counterfactual result. The model includes Oregon as the treatment group and Idaho as the control. These states were chosen because of their similarities in proximity, culture (especially between eastern Oregon and western Idaho), terrain, and economy. Whatever differences between the two states which were unobservable via monthly data are captured by the additional entity fixed effects included within the model. Fixed effects are a statistical method which helps to capture unobservable variables and are represented through a binary operator within the model, in this case the variable OR. They capture the unobservable variables, such as any other differences in culture and geography between the two states, that remain constant over time. With entity fixed effects, we are better able to estimate the difference in policy outcome from the ballot measure by separating out unobserved factors that may be playing a role in its outcome, crime rate. The model specification is as follows:

$$C_i = \beta_1 T_i + \beta_2 P_i + \beta_3 TxP + \alpha_i + w_i\gamma + u_i$$

Where C_i is the dependent variable crime rate, for which crime rate is per 100 thousand people. β_1 is the treatment group specific effect that accounts for average permanent differences between the states, β_2 is the time trend, and β_3 is the treatment effect. (variable of interest). The treatment effect is given by the interaction between the Treatment and OR variables. α_i captures the entity fixed effects between Oregon and Idaho, and w represents the added control variables. The controls included relate to crime rate such as educational attainment, unemployment rate, and weather. Lastly, u_i is the error term for this model.

Outline/Analysis Plan:

For my analysis, I will run three different models with differing levels of complexity in order to address my research question of how the introduction of Ballot Measure 110 has affected crime rate for that state of Oregon. The hypothesis, therein, will be to specifically test whether or not Ballot Measure 110 had no effect on crime rate. To test this hypothesis, I will be using the data I collected in the differing models to estimate the effect the independent variable, treatment effect (an interacted term between Treatment and OR), on the dependent variable crime rate. Each model will contain additional covariates to see how the effect on crime rate changes with each iteration of differing levels of control. Additionally, an F-test will be run to test for joint significance of covariates. The final model specification, however, will be dependent on additional covariates and whether they are important to theorical estimation of crime rate. These include the aforementioned socio-economic factors such as education level and unemployment rates, but also variables related to the seasonality in crime rate. Finally, to test the hypothesis, we will use t-test to see if our estimated effect is significantly different from zero. Then, upon its statistical significance evaluation, we can interpret our results to see what economic significance it may have.

Results:

To test our hypothesis:

$$H_0: \beta_3 = 0 \;\; H_1: \beta_3 \neq 0$$

I ran multiple models in order to make an informed judgement on our final determination of our hypothesis question. Before I could definitively say which specification was best statistically, I

needed to conduct an F-Test to see which covariates were significant to explaining my dependent variable crime rate. Below are the results of the tests.

Table 3: F-stat Variables Test First Test: With Month Variables	 Second Test: Without Month Variables
(1) Treatment = 0 (2) OR = 0 (3) Unemployment Rate = 0 (4) Percent college = 0 (5) February = 0 (6) March = 0 (7) April = 0 (8) May = 0 (9) June = 0 (10) July = 0 (11) August = 0 (12) September = 0 (13) October = 0 (14) November = 0 (15) December = 0 chi2(1) = 1.41 Prob > chi2 = 0.2359	(1) Treatment = 0 (2) OR = 0 (3) Unemployment Rate = 0 (4) Percent college = 0 chi2(1) = 248.24 Prob > chi2 = 0.0000

From the F-stat table, we see that the addition of monthly dummy variables results in the p-value associated with the F-stat to fall below statistically significant threshold. When conducting a new test based off of all other variables, we see that the test becomes significant. The results of the F-test, however, do not necessarily override the expectations from our theoretical foundations. If economic theory informs us that certain variables should have effect on our dependent variable. Therefore, for the final model specification, the month dummy variables are included to control for such factors.

Looking at our final results in the regression table below, we see significant results in both the time trend and our variable of interest, treatment effect.

Table 4:
Regression Results

	(1) Crime Rate	(2) Crime Rate	**(3) Crime Rate**
Treatment Group specific effect	-32.90***	11.61	**-23.62**
	(1.39e-12)	(19.65)	**(24.17)**
Time Trend	479.3***	426.6***	**462.7*****
	(1.00e-12)	(25.06)	**(29.36)**
Treatment Effect	-21.29***	-29.11***	**-30.51*****
	(1.46e-12)	(5.803)	**(7.196)**
Unemployment Rate		-5.721	**-12.23**
		(5.792)	**(7.574)**
Percentage College		13.14	**8.625**
		(7.567)	**(9.205)**
February			**-44.42**
			(37.47)
March			**21.47*****
			(5.967)
April			**40.39*****
			(0.430)
May			**60.71*****
			(8.527)
June			**45.16*****
			(5.704)
July			**69.70*****
			(11.60)
August			**65.88*****
			(8.450)
September			**44.71*****
			(7.618)
October			**52.10*****
			(10.73)
November			**5.240**
			(18.22)
December			**-8.070**
			(18.29)
_cons	246.2***	-384.1	**-148.9**
	(9.53e-13)	(349.5)	**(418.2)**
N	76	76	**75**
adj. R^2			

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

This means that the difference in crime rate of the treatment group (Oregon) before and after the ballot measure, when accounting for the difference in crime rate of the control group (Idaho) before and after the treatment is significantly different from zero. It is also important to note that the sign of the coefficient is negative, rather than the hypothesized result of positive. Therefore, we can say the results of the ballot measure actually reduced crime relative to the time period before the law was introduced.

Looking at the economic implications of the results, we see that the ballot measure reduced crime rate (per 100k) by around 31. On average, through the three years of observations, Oregon had a crime rate of 689.89. This represents around a 4% decrease in overall crime rate from the ballot measure's introduction. Economically speaking, this is a sizeable decrease for the state. Moreover, this is another piece of evidence that the conventional thinking surrounding drug decriminalization doesn't hold up in empirical research on the subject.

Conclusion:

Based on our results, we found a counter example to the widely held belief that drug decriminalization leads to higher levels of crime. It appears that Oregon's crime rate decreased post-treatment relative to our control Idaho. While other evidence is limited, another study by non-profit RTI International found similar results when comparing 911 calls across cities in the pacific northwest before and after the new law went into effect (RTI, 2022). That is not to say, however, that the results presented here are without their issues. Firstly, in future research, time fixed effects need to be included to account for time variant factors that are consistent across these states. Likewise, certain factors need to be better controlled for in future studies or follow-ups. The pandemic for instance could have played a significant role in either quelling crime or leading to a

larger outburst as the pandemic eased. Alternatively, our treatment (Oregon) could have been influenced by the widescale protests in 2020 surrounding police brutality and the George Floyd murder case, thus leading to larger levels of crime in pre-treatment time periods. Bigger cities, such as Portland, also lack a direct comparison in our control state of Idaho, leading to the breakdown of our assumption of similar characteristics between our treatment and control groups. Follow up research may only include more comparable counties in each state (such as in eastern Oregon and western Idaho). Lastly, more data can be included in the pre and post time periods to understand long-term effects of such a change in drug policy. Nevertheless, despite its issues, these results offer promising insights into the possibility of future drug decriminalization. With the potential of new laws like this to spread to other states and localities, as marijuana legalization had done before it, relevant research on drug decriminalization and ballot measure 110 will become all the more crucial in informing policymakers, public health experts, and the general public about the impacts of such measures. If drug decriminalization can be shown to have a negative relationship with crime rates, as my research indicates, this provides a strong justification for further implementation elsewhere. The potential justification for this new approach is furthered considering the many issues associated with traditional "war on drug" policies of old, such as racial inequity and prison overabundance. More broadly, if drug decriminalization can be shown to have a negative relationship with crime rate, there could be additional benefits associated with the policy which could be highlighted with further analysis, such as with diversion programs and access to treatment facilities.

Overall, the significance of this research lies in its potential to inform stakeholders on a more evidence-based approach toward drug policy. One that would have benefits toward public-

safety, equality, and humanity when interacting with some of the most vulnerable people within our communities.

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